```
-- ========================================
-- OJT MANAGEMENT SYSTEM
-- PART 1: TABLE CREATION
-- ========================================

CREATE TABLE users (
    user_id SERIAL PRIMARY KEY,
    username VARCHAR(50) NOT NULL UNIQUE,
    email VARCHAR(255) NOT NULL UNIQUE,
    password_hash VARCHAR(255) NOT NULL,
    role VARCHAR(30) NOT NULL
        CHECK (role IN (
            'ADMIN',
            'STUDENT',
            'ACADEMIC_SUPERVISOR',
            'COMPANY_SUPERVISOR',
            'OJT_COORDINATOR'
        )),
    is_active BOOLEAN NOT NULL DEFAULT TRUE,
    created_at TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP,
    updated_at TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP,
    deleted_at TIMESTAMP NULL
);

CREATE TABLE audit_log (
    audit_id SERIAL PRIMARY KEY,
    table_name VARCHAR(100) NOT NULL,
    action VARCHAR(30) NOT NULL,
    row_id INTEGER NOT NULL,
    action_time TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP,
    details JSONB NULL
);

CREATE TABLE students (
    student_id SERIAL PRIMARY KEY,
    user_id INTEGER NOT NULL UNIQUE
        REFERENCES users(user_id),
    student_number VARCHAR(30) NOT NULL UNIQUE,
    first_name VARCHAR(50) NOT NULL,
    last_name VARCHAR(50) NOT NULL,
    middle_name VARCHAR(50) NULL,
    course VARCHAR(100) NOT NULL,
    year_level INTEGER NOT NULL
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        CHECK (year_level > 0),
    contact_number VARCHAR(20) NOT NULL,
    created_at TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP,
    deleted_at TIMESTAMP NULL
);

CREATE TABLE companies (
    company_id SERIAL PRIMARY KEY,
    company_name VARCHAR(255) NOT NULL UNIQUE,
    industry VARCHAR(100) NOT NULL,
    address VARCHAR(255) NOT NULL,
    contact_number VARCHAR(20) NOT NULL,
    email VARCHAR(255) NOT NULL,
    is_partner BOOLEAN NOT NULL DEFAULT FALSE,
    created_at TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP,
    deleted_at TIMESTAMP NULL
);

CREATE TABLE academic_supervisors (
    academic_sup_id SERIAL PRIMARY KEY,
    user_id INTEGER NOT NULL UNIQUE
        REFERENCES users(user_id),
    first_name VARCHAR(50) NOT NULL,
    last_name VARCHAR(50) NOT NULL,
    created_at TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP,
    deleted_at TIMESTAMP NULL
);

CREATE TABLE company_supervisors (
    company_sup_id SERIAL PRIMARY KEY,
    user_id INTEGER NOT NULL UNIQUE
        REFERENCES users(user_id),
    company_id INTEGER NOT NULL
        REFERENCES companies(company_id),
    first_name VARCHAR(50) NOT NULL,
    last_name VARCHAR(50) NOT NULL,
    position VARCHAR(100) NOT NULL,
    contact_number VARCHAR(20) NOT NULL,
    created_at TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP,
    deleted_at TIMESTAMP NULL
);

CREATE TABLE requirements (
    req_id SERIAL PRIMARY KEY,
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    title VARCHAR(150) NOT NULL UNIQUE,
    description TEXT NULL,
    is_mandatory BOOLEAN NOT NULL DEFAULT TRUE,
    created_at TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP
);


CREATE TABLE ojt_applications (
    application_id SERIAL PRIMARY KEY,
    student_id INTEGER NOT NULL REFERENCES students(student_id),
    company_id INTEGER NOT NULL REFERENCES companies(company_id),
    application_date DATE NOT NULL DEFAULT CURRENT_DATE,
    status VARCHAR(20) NOT NULL DEFAULT 'Pending'
        CHECK (status IN ('Pending', 'Approved', 'Rejected', 'Withdrawn')),
    remarks TEXT NULL,
    created_at TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP,
    deleted_at TIMESTAMP NULL,

    -- Added composite unique constraint
    UNIQUE (application_id, student_id, company_id)
);



);

CREATE TABLE student_requirements (
    student_req_id SERIAL PRIMARY KEY,
    student_id INTEGER NOT NULL
        REFERENCES students(student_id),
    req_id INTEGER NOT NULL
        REFERENCES requirements(req_id),
    verified_by_sup_id INTEGER NULL
        REFERENCES academic_supervisors(academic_sup_id),
    file_path VARCHAR(255) NULL,
    status VARCHAR(20) NOT NULL DEFAULT 'Pending'
        CHECK (status IN (
            'Pending', 'Submitted', 'Approved', 'Rejected'
        )),
    submission_date TIMESTAMP NULL,
    verified_at TIMESTAMP NULL,
    remarks TEXT NULL,
    UNIQUE (student_id, req_id)
);
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CREATE TABLE ojt_placements (
    placement_id SERIAL PRIMARY KEY,
    application_id INTEGER NOT NULL UNIQUE,
    student_id INTEGER NOT NULL,
    company_id INTEGER NOT NULL,
    company_sup_id INTEGER NOT NULL
        REFERENCES company_supervisors(company_sup_id),
    academic_sup_id INTEGER NOT NULL
        REFERENCES academic_supervisors(academic_sup_id),
    start_date DATE NOT NULL,
    end_date DATE NOT NULL,
    required_hours NUMERIC(6,2) NOT NULL
        CHECK (required_hours > 0),
    status VARCHAR(20) NOT NULL DEFAULT 'Pending'
        CHECK (status IN ('Pending', 'Active', 'Completed', 'Terminated')),
    created_at TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP,
    deleted_at TIMESTAMP NULL,

    -- Added composite foreign key
    CONSTRAINT fk_placement_application_consistency
        FOREIGN KEY (application_id, student_id, company_id)
        REFERENCES ojt_applications
            (application_id, student_id, company_id),

    CHECK (start_date <= end_date)
);



CREATE TABLE attendance (
    attendance_id SERIAL PRIMARY KEY,
    placement_id INTEGER NOT NULL
        REFERENCES ojt_placements(placement_id),
    attendance_date DATE NOT NULL,
    time_in TIME NULL,
    time_out TIME NULL,
    hours_rendered NUMERIC(5,2) NOT NULL DEFAULT 0
        CHECK (hours_rendered >= 0),
    is_verified BOOLEAN NOT NULL DEFAULT FALSE,
    created_at TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP,
    UNIQUE (placement_id, attendance_date),
    CHECK (
        time_in IS NULL OR time_out IS NULL
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        OR time_out >= time_in
    )
);

CREATE TABLE evaluations (
    evaluation_id SERIAL PRIMARY KEY,
    placement_id INTEGER NOT NULL
        REFERENCES ojt_placements(placement_id),
    company_sup_id INTEGER NOT NULL
        REFERENCES company_supervisors(company_sup_id),
    score NUMERIC(5,2) NOT NULL
        CHECK (score >= 0 AND score <= 100),
    feedback TEXT NULL,
    evaluation_date DATE NOT NULL DEFAULT CURRENT_DATE,
    created_at TIMESTAMP NOT NULL DEFAULT CURRENT_TIMESTAMP
);



-- ========================================
-- PART 2: POSTGRESQL LOGIN ROLE CREATION
-- Run as a database administrator.
-- ========================================

CREATE ROLE ojt_admin LOGIN;
CREATE ROLE ojt_student LOGIN;
CREATE ROLE ojt_academic_supervisor LOGIN;
CREATE ROLE ojt_company_supervisor LOGIN;
CREATE ROLE ojt_coordinator LOGIN;

-- Database connection access
GRANT CONNECT ON DATABASE ojt_management TO
    ojt_admin,
    ojt_student,
    ojt_academic_supervisor,
    ojt_company_supervisor,
    ojt_coordinator;

-- Schema access
GRANT USAGE ON SCHEMA public TO
    ojt_admin,
    ojt_student,
    ojt_academic_supervisor,
    ojt_company_supervisor,
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    ojt_coordinator;

-- ADMIN
GRANT SELECT, INSERT, UPDATE ON users TO ojt_admin;
GRANT SELECT ON audit_log TO ojt_admin;

-- STUDENT
GRANT SELECT ON
    students,
    companies,
    requirements,
    ojt_applications,
    student_requirements,
    ojt_placements,
    attendance,
    evaluations
TO ojt_student;

GRANT INSERT, UPDATE ON
    ojt_applications,
    student_requirements,
    attendance
TO ojt_student;

-- ACADEMIC SUPERVISOR
GRANT SELECT ON
    students,
    ojt_applications,
    student_requirements,
    ojt_placements,
    attendance,
    evaluations,
    requirements
TO ojt_academic_supervisor;

GRANT UPDATE (
    verified_by_sup_id,
    status,
    verified_at,
    remarks
) ON student_requirements
TO ojt_academic_supervisor;

GRANT UPDATE (status) ON ojt_placements
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TO ojt_academic_supervisor;

GRANT UPDATE (is_verified) ON attendance
TO ojt_academic_supervisor;

-- COMPANY SUPERVISOR
GRANT SELECT ON
    students,
    companies,
    company_supervisors,
    ojt_placements,
    attendance,
    evaluations
TO ojt_company_supervisor;

GRANT INSERT, UPDATE ON attendance
TO ojt_company_supervisor;

GRANT INSERT, UPDATE ON evaluations
TO ojt_company_supervisor;

-- OJT COORDINATOR
GRANT SELECT ON
    students,
    companies,
    academic_supervisors,
    company_supervisors,
    ojt_applications,
    student_requirements,
    ojt_placements,
    attendance,
    evaluations,
    requirements
TO ojt_coordinator;

GRANT INSERT, UPDATE ON
    companies,
    company_supervisors,
    ojt_applications,
    ojt_placements
TO ojt_coordinator;
```

```
-- ADMIN: create user accounts
GRANT USAGE ON SEQUENCE users_user_id_seq
TO ojt_admin;

-- STUDENT: create applications, requirements, attendance
GRANT USAGE ON SEQUENCE
    ojt_applications_application_id_seq,
    student_requirements_student_req_id_seq,
    attendance_attendance_id_seq
TO ojt_student;

-- COMPANY SUPERVISOR: create attendance and evaluations
GRANT USAGE ON SEQUENCE
    attendance_attendance_id_seq,
    evaluations_evaluation_id_seq
TO ojt_company_supervisor;

-- COORDINATOR: create companies, supervisors,
-- applications, and placements
GRANT USAGE ON SEQUENCE
    companies_company_id_seq,
    company_supervisors_company_sup_id_seq,
    ojt_applications_application_id_seq,
    ojt_placements_placement_id_seq
TO ojt_coordinator;




-- sample application account per role
-- The placeholder is not a valid authentication hash.

INSERT INTO users (
    username,
    email,
    password_hash,
    role
```

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)
VALUES
    ('admin_account', 'admin@example.com',
     'REPLACE_WITH_SECURE_PASSWORD_HASH', 'ADMIN'),

    ('student_account', 'student@example.com',
     'REPLACE_WITH_SECURE_PASSWORD_HASH', 'STUDENT'),

    ('academic_account', 'academic@example.com',
     'REPLACE_WITH_SECURE_PASSWORD_HASH', 'ACADEMIC_SUPERVISOR'),

    ('company_account', 'company@example.com',
     'REPLACE_WITH_SECURE_PASSWORD_HASH', 'COMPANY_SUPERVISOR'),

    ('coordinator_account', 'coordinator@example.com',
     'REPLACE_WITH_SECURE_PASSWORD_HASH', 'OJT_COORDINATOR');
```